Endogenous Education Choice in Segmented Search Markets

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Abstract

How do workers decide whether to acquire education? The standard answer weighs wages against costs, but this misses a deeper aspect of the choice: education is not just about whether the skilled market pays more, it is about whether that market will actually have jobs available after training that were worth the trouble. This paper develops a general equilibrium search model in which the fraction of workers who choose to train is determined endogenously alongside wages, unemployment, and market tightness in both markets. Workers differ in ability, which governs the cost of training; the skilled market additionally features on-the-job search and a job ladder, giving trained workers richer career dynamics than their unskilled counterparts. The model is estimated on U.S. data across multiple aggregate states of the economy using Simulated Method of Moments. This allows me to trace how the economy — and the education margin — adjusts over time after an aggregate shock. I follow this with two applications that demonstrate the model's usefulness: the first studies the effect of more generous outside options (UBI or unemployment insurance) on unemployment duration and match quality; the second compares two education subsidy designs: flow transfers during training versus upfront cost reductions.

1 Introduction

College enrolment rises during recessions. This is one of the most robust findings in the economics of education: when labour markets deteriorate, workers respond not only by searching harder for jobs but also by investing in human capital (Dellas and Sakellaris, 2003; Clark, 2011; Barr and Turner, 2015). Aggregate skill composition, by contrast, adjusts slowly after major crises—the skilled workforce share barely moved in the two years following the 2008 financial crisis, even as unemployment spiked. These two facts sit in tension. If training decisions respond to cyclical conditions, why does the aggregate skill distribution move so little? And why did the financial crisis and COVID-19 appear to generate qualitatively different enrolment responses?

The standard account treats training as a productivity-upgrading decision: a worker compares the wage return to skills against opportunity costs. This framing misses something important. Investing in training is not just a choice about productivity—it is a choice about *which labour market to participate in for the rest of one's career*. Unskilled and skilled labour markets are governed by qualitatively different dynamics. In the unskilled market, matches start at frontier quality and can only be damaged by idiosyncratic shocks; a worker's only avenue for improvement is to re-enter unemployment and draw a new match. In the skilled market, workers search on the job, receive outside offers, and climb a wage ladder over time. The relevant comparison for a worker contemplating training is not a marginal wage return against a marginal cost—it is the entire continuation value of the skilled market, including its job-ladder dynamics, against the entire continuation value of remaining unskilled. Both continuation values depend on equilibrium conditions in both markets simultaneously.

This paper asks: what determines the equilibrium training threshold in a frictional labour market, and how does it respond to aggregate shocks and policy?

To answer this, I develop a structural equilibrium model in which two frictional labour markets coexist and workers endogenously choose which to enter. The unskilled market follows a standard Diamond (1982)–Mortensen and Pissarides (1994)–Pissarides (2000) framework with endogenous match destruction: matches begin at frontier quality and are damaged by Poisson productivity shocks, with workers and firms separating endogenously when productivity falls below a reservation threshold. The skilled market additionally features on-the-job search: employed workers pay a flow cost and receive outside offers, climbing a job ladder à la Cahuc et al. (2006). Training grants permanent access to the skilled market at a one-time ability-dependent cost. The training threshold—the ability level at which a worker is indifferent between the two markets—is an equilibrium object determined jointly with the

compositions of both unemployment pools and market tightness in both sectors.

The key consequence of this joint determination is a general-equilibrium feedback that existing models miss. The mass and composition of the skilled pool feed into market tightness through firms' free-entry conditions, which in turn feed back into the value of entering the skilled market. A shock or policy that shifts the training threshold therefore triggers responses in both markets simultaneously: in the unskilled market, fewer searchers mean tighter conditions and better outside options, damping the direct effect; in the skilled market, more entrants mean more competition for jobs, compressing the value of the skilled continuation. A model that holds either market fixed misses these feedbacks entirely.

The model is taken to U.S. labour-market data using the Simulated Method of Moments across four distinct aggregate regimes: a pre-crisis baseline (2003–07), the financial crisis (2008–09), a pre-COVID expansion (2015–19), and the COVID shock (2020–21). The estimation targets 23 moments—unemployment rates, transition rates, wage distributions, and market tightness by skill group—within each window. A multi-regime estimation strategy separates *deep structural parameters*—governing technology and institutions, invariant to the business cycle—from *regime-specific parameters* that capture aggregate conditions. Deep parameters are estimated once on the pre-crisis baseline and held fixed across regimes, providing both a structural stability check and a disciplined interpretation of each shock.

The model's structure generates qualitatively different predictions for the two crises. A symmetric demand shock—in which both markets deteriorate roughly equally—raises the option value of waiting and pushes the training threshold up, so that fewer workers find it optimal to train. An asymmetric supply-and-reallocation shock that hits the unskilled sector disproportionately, as COVID did through contact-intensive services, compresses the continuation value of remaining unskilled while expanded unemployment benefits (FPUC, PUA) raise the flow payoff during the training spell. The training threshold falls, and enrolment rises—a structural mechanism consistent with the enrolment spike documented by Barr and Turner (2015) and Blom et al. (2021). Because both predictions emerge from the same model with the same deep parameters—only the regime-specific parameters change—the framework can in principle decompose which aspects of each crisis drive the sign reversal.

The policy exercise evaluates two education-subsidy designs. Policy A raises the flow payoff during the training spell b_T—monthly living-cost grants that make the training period less costly on a flow basis. Policy B reduces the one-time training cost $c(x)$ by replacing the cost coefficient a_c with $(1-\delta)\,a_c$—an investment in higher-education quality that lowers the upfront hurdle. Both policies shift the training threshold, but through different channels

and with different general-equilibrium feedbacks. Whether flow subsidies or cost-reduction subsidies are more effective per unit of fiscal cost depends on which margin of the training decision is more responsive—a question that requires the full general-equilibrium framework to answer.

In sum, the paper develops the first search-and-matching framework in which the boundary between two coexisting frictional labour markets is itself an equilibrium object, jointly determined with market tightness and pool composition in both sectors through free-entry conditions. The estimation strategy—multi-regime SMM with a structural stability check across baselines—provides a disciplined way to identify what changed during each crisis. The policy exercise evaluates competing education subsidies, quantifying a general-equilibrium tightness channel that competitive models miss (Abbott et al., 2019).

The rest of the paper is organised as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 describes the data, moments, identification, and estimation strategy. Section 5 reports the estimation results, applies the framework to the financial crisis and COVID-19, and compares the two episodes. Section 6 evaluates the education-subsidy policies. Section 7 concludes.

2 Related Literature

This paper connects five bodies of work.

Education choice in search models

The most direct predecessors are the papers that embed education decisions in search-and-matching frameworks. Charlot and Decreuse (2005) build a two-sector model in which workers self-select into educated and uneducated pools; both markets clear simultaneously, and a composition externality arises because the marginal entrant degrades the mean ability of the pool she joins, depressing vacancy creation for all co-searchers. Charlot et al. (2005) study how education raises both adaptability and productivity in a single frictional market. Charlot and Decreuse (2010) extend the two-sector framework to two-dimensional heterogeneity—ability and schooling cost—and show that the rich over-invest in education while the poor under-invest. Charlot and Malherbet (2013) add employment protection and show that a direct training subsidy can correct the hold-up distortion that search frictions create for education investments. Wasmer (2006) studies the choice between general and specific skills in a single Mortensen–Pissarides market with firing costs.

These papers share a common limitation: the skilled market has no internal career dynamics. Once a worker enters the educated pool, wages are set by Nash bargaining and the match persists until exogenous destruction. There is no on-the-job search, no job ladder, and no mechanism through which trained workers climb toward better matches over time. The present paper adds exactly this structure. The skilled market features on-the-job search and a job ladder à la Cahuc et al. (2006), so that the value of training depends not only on the entry wage but on the entire trajectory of wage gains from climbing the ladder. This interaction between the education margin and job-ladder dynamics is absent from the existing education-in-search literature.

Search and matching foundations

The model builds on the search-and-matching tradition of Diamond (1982) and Pissarides (2000), with endogenous job destruction from Mortensen and Pissarides (1994) governing match dissolution in the unskilled sector. The job-ladder dynamics in the skilled sector rest on the framework of Postel-Vinay and Robin (2002), who show how on-the-job search and between-employer competition generate equilibrium wage dispersion. Cahuc et al. (2006) embed Nash bargaining into this environment—giving workers positive bargaining power while retaining the job-ladder structure—and provide the wage-determination protocol that I adopt.

Robin (2011) extends the job-ladder model to include aggregate shocks and shows that endogenous job destruction coupled with worker heterogeneity amplifies productivity shocks, a mechanism that also operates in my model and generates countercyclical training incentives. Lise et al. (2016) and Lise and Robin (2017) introduce continuous worker and firm heterogeneity and study how sorting between the two sides evolves over the business cycle, but they take the type distribution in each market as exogenous. My paper endogenises it: the distribution of ability types among skilled workers is an equilibrium object, determined by the training cutoff.

Acemoglu (2001) studies a related mechanism in a different setting: when good and bad jobs coexist in a single search market, higher unemployment benefits shift the composition of employment toward good jobs by raising workers' reservation values. My model extends this logic to a two-market environment where the composition shift operates through the training margin rather than through firms' job-creation decisions within a single market.

Human capital in search models

A separate literature studies training and skill accumulation within a given search market, without the market-selection margin. Acemoglu and Shimer (1999) establish that search frictions create a hold-up problem for pre-market human capital investments: because wages are determined ex post, workers underinvest in education relative to the social optimum. This underinvestment result—confirmed in a richer environment by Charlot and Malherbet (2013)—motivates the policy exercises in Section 6, where subsidies to training can correct the frictional distortion. Bagger et al. (2014) allow human capital to accumulate along the job ladder, and Audoly et al. (2024) build on this to quantify the contributions of job-ladder losses, skill depreciation, and firm-specific capital to the earnings losses of displaced workers. In these papers, training unfolds within a single frictional environment: it raises the worker's productivity or bargaining position but does not change which market the worker participates in.

This paper takes a different view. Training is a one-time, costly decision that grants permanent access to a qualitatively different labour market. The worker compares two entire continuation value functions, each governed by different search frictions, different turnover dynamics, and different equilibrium objects. This makes the training decision fundamentally more sensitive to general-equilibrium conditions than the within-match view implies. The model abstracts from within-market skill accumulation by treating ability x as a fixed type. A natural extension would allow a skill multiplier to evolve with labour-market experience; Cohen et al. (2025) document that unemployment spells erode measurable skills independently of ability, providing empirical motivation for such an extension.

Education policy in general equilibrium

The competitive general-equilibrium literature on education policy—notably Heckman et al. (1998) and Abbott et al. (2019)—evaluates subsidies in OLG frameworks with competitive factor markets. In those models, the response to a subsidy operates through the skill premium: more educated workers raise relative supply and compress wages. Bárány (2016) adds endogenous directed technical change to this logic, showing that the education margin interacts with the direction of innovation. In all of these papers the relevant general-equilibrium channel is the relative price of skill. In my model, the channel is different. More trainees enter skilled unemployment, which affects the job-finding rate of all skilled unemployed workers by changing market tightness. This tightness feedback can amplify or dampen the direct effect of a subsidy in ways that competitive models cannot capture.

The policy exercise distinguishes two subsidy instruments: a flow transfer during the training spell and a one-time reduction in training costs. This distinction maps onto a design dimension studied empirically by Lindner and Reizer (2020), who show that the timing of benefit payments—flow versus frontloaded—generates different search responses holding total transfers constant. The exercise is also connected to the active labour market programme evaluation literature: Card et al. (2018) synthesise over 200 programme evaluations and find that training programmes generate larger positive effects in the medium run than in the short run, a pattern consistent with the job-ladder dynamics in the skilled sector, where newly trained workers enter at the bottom of the quality distribution and climb over time.

The empirical finding of Nekoei and Weber (2017) that more generous unemployment benefits improve reemployment match quality—not just search duration—provides the reduced-form motivation for the role of non-employment payoffs in the model: a higher b_U raises reservation match quality in the unskilled market, but it also compresses the net gain from training and shifts the training threshold, generating cross-market effects that a single-market framework cannot capture.

Countercyclical enrolment

A parallel empirical literature documents that college enrolment is countercyclical. Dellas and Sakellaris (2003) establish this pattern using four decades of US data. Clark (2011) confirms it for England using regional variation in youth unemployment. Barr and Turner (2015) show that the Great Recession triggered enrolment increases concentrated at non-selective institutions, and that unemployment-insurance benefit duration played a significant role in the timing of these decisions. Blom et al. (2021) find that community college enrolment responds to local labour-market conditions with an elasticity of approximately 0.1; Ulvestad and Skjelbred (2023) extend the evidence to graduate-level enrolment in Norway, confirming that the pattern is not specific to U.S. undergraduates. The model generates countercyclical enrolment endogenously: a recession lowers the contact rate in the unskilled market, compressing the continuation value of remaining unskilled and lowering the training threshold $\bar{x}(z)$, so that a larger mass of workers finds it optimal to train. This is the structural mechanism behind the aggregate enrolment patterns that the empirical literature documents.

Bičáková et al. (2021) and Bičáková et al. (2023) go further and show that graduates who enrol during recessions earn higher wages later, an effect that cannot be attributed to ability selection at entry. Their finding of a conditional wage premium operating through effort and preference formation during college lies outside the scope of the current model, which abstracts from the training spell itself.

3 Model

I develop a general-equilibrium search model in which workers endogenously choose whether to invest in training and thereby gain access to a higher-skill labour market. Training decisions, wages, unemployment, and market tightness are determined jointly in both markets, so the education margin responds not only to the skilled wage premium but also to the availability of skilled jobs after training. The education-choice structure draws on Charlot and Decreuse (2005, 2010), while the skilled-market dynamics build on the job-ladder framework of Postel-Vinay and Robin (2002) with the Nash-bargaining wage protocol of Cahuc et al. (2006).

3.1 Environment

Time is continuous and all agents discount the future at rate $r > 0$. A unit mass of infinitely lived workers, each characterised by an innate type $x \in [0, 1]$ drawn from a Beta(a_ℓ, b_ℓ) distribution with CDF L and density ℓ, populates the economy. Workers exit at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers drawn from the same distribution.

Workers participate in one of two labour markets—unskilled (U) or skilled (S)—and can move from the first to the second by completing training. In both markets firms are homogeneous and jobs are characterised by an idiosyncratic match-specific productivity $p \in (0, 1]$. The key asymmetry is in how match quality evolves: in the unskilled market, new matches begin at the frontier quality $p_0 = 1$ and deteriorate after shocks, whereas in the skilled market quality is drawn from a distribution $\Gamma(\cdot \mid z)$ at formation and redrawn after shocks. The skilled market also supports on-the-job search and a job ladder, giving trained workers richer career dynamics.

Aggregate states

The aggregate state of the economy is $z \in \mathcal{Z}$, where $\mathcal{Z}$ is a finite set. A regime z determines not merely sector productivities but the full constellation of parameters governing each market:

$$z \longmapsto \big\{P_U(z),\, P_S(z),\, \alpha_U(z),\, a_\Gamma(z),\, b_\Gamma(z),\, \lambda_U(z),\, \lambda_S(z),\, \xi_S(z),\, k_U(z),\, k_S(z)\big\}. \tag{1}$$

The sector-specific productivities $P_j(z)$ scale output, the parameters $\alpha_U(z)$, $a_\Gamma(z)$, and $b_\Gamma(z)$ govern the match-quality distributions, $\lambda_j(z)$ are shock arrival rates, $\xi_S(z)$ is the exogenous

skilled separation hazard, and $k_j(z)$ are vacancy-posting costs. When the economy switches from one regime to another, all of these objects shift jointly.

At an exogenous date T, the economy switches unexpectedly and permanently from an initial regime z_0 to a new regime $z_1 \neq z_0$.

3.2 Production

A match between a worker of type x and a job with match-specific productivity p in sector $j \in \{U, S\}$ produces flow output

$$\pi_j(x, p, z) = P_j(z)\, x\, p. \tag{2}$$

Output is multiplicatively separable in ability and match quality. The sector shifter $P_j(z)$ captures aggregate conditions. Training changes market access only: it does not raise innate productivity or generate within-market human capital accumulation.

3.3 Matching

Each market has its own Cobb–Douglas matching function

$$M_j\big(U_j^{\text{eff}}, V_j\big) = \mu_j \left(U_j^{\text{eff}}\right)^{\eta_j} V_j^{1-\eta_j}, \qquad \eta_j \in (0, 1). \tag{3}$$

In the unskilled market only unemployed workers search, so $U_U^{\text{eff}} = U_U$. In the skilled market employed workers may also search on the job, so the effective pool of seekers is $U_S^{\text{eff}}(z) = U_S(z) + \int\!\!\int s^*(x, p, z)\, e_S(x, p, z)\, dp\, dx$, where $s^* \in \{0, 1\}$ is the on-the-job search decision and e_S is the density of skilled employed. Market tightness is $\theta_j(z) = V_j(z)/U_j^{\text{eff}}(z)$, the vacancy-filling rate is $q_j(\theta_j) = \mu_j \theta_j^{-\eta_j}$, and the job-finding rate for a seeker is $\theta_j(z)\, q_j(\theta_j(z))$. Throughout I write $\kappa_j(z) \equiv \theta_j(z)\, q_j(\theta_j(z))$ for the contact rate in market j: the rate at which any active job seeker—whether unemployed or searching on the job—meets a vacancy. These contact rates are the key prices through which labour-market conditions feed into training decisions.

3.4 Training

Only untrained workers who are currently unemployed in the unskilled market may start training. Doing so requires a one-time utility cost $c(x) = c_0\,(1 - x)\, e^{-x}$ that declines in ability. Once a worker begins training, completion arrives at Poisson rate $\phi > 0$, at which point the worker enters skilled unemployment.

While out of work, workers receive flow payoffs that depend on their status: b_U for unskilled unemployed, b_T for those in training, and b_S for skilled unemployed. These payoffs represent the value of non-employment to the worker—a composite of home production, informal income, and the amenity value of leisure. They are structural parameters, constant across aggregate states.[1]

The value of being in training satisfies

$$(r + \phi + \nu)\, T(x, z) = b_T + \phi\, U_S(x, z), \tag{4}$$

so it depends on the value of skilled unemployment $U_S(x, z)$—the channel through which skilled-market conditions feed into the training decision.

An entering unskilled unemployed worker of type x faces the choice between searching in the unskilled market and paying the training cost to begin the pipeline toward the skilled market. The value of searching satisfies $(r + \nu)\, U_U^{\text{search}}(x, z) = b_U + \kappa_U(z)[E_U(x, 1, z) - U_U^{\text{search}}(x, z)]$, reflecting the flow payoff b_U and the option of contacting a vacancy at the frontier quality $p_0 = 1$. The overall value of unskilled unemployment is then

$$U_U(x, z) = \max\left\{U_U^{\text{search}}(x, z),\ -c(x) + T(x, z)\right\}. \tag{5}$$

Because $c'(x) < 0$ and $T(x, z)$ is increasing in x, the net gain from training is increasing in ability. This yields a cutoff.

Proposition 1 (Training cutoff)**.** *Fix a regime $z \in \mathcal{Z}$. There exists a threshold $\bar{x}(z) \in [0, 1]$ such that the optimal training policy is $\tau(x, z) = \mathbf{1}\{x > \bar{x}(z)\}$.*

The cutoff $\bar{x}(z)$ summarises the education margin. It responds to aggregate conditions through three channels: the skilled wage premium, the probability of finding a skilled job after training, and the value of the unskilled outside option.

[1] In the U.S., UI eligibility requires prior employment and expires after a fixed duration, so b_j should not be read literally as an unemployment-insurance payment. Rather, b_j captures the average flow value that an unemployed worker in segment j enjoys, inclusive of any UI entitlements, home production, and other non-market income. In the policy experiments of Section 6 I vary b_U and b_S jointly to study the effect of a permanent increase in the generosity of the non-employment safety net.

3.5 Unskilled Labour Market

Match quality and shocks

New unskilled matches begin at the frontier quality $p_0 = 1$, so every hire starts at full productivity. Idiosyncratic damage shocks arrive at Poisson rate $\lambda_U(z)$, after which a new quality is drawn from $G_z(p) = p^{\alpha_U(z)}$. If the new draw falls below a reservation threshold $p_U^*(x, z)$—to be determined in equilibrium—the match is destroyed and the worker returns to unemployment.

Surplus

Nash bargaining with worker share β_U splits the total surplus $S_U(x, p, z) = [E_U(x, p, z) - U_U(x, z)] + J_U(x, p, z)$ in fixed proportions, where E_U is the worker's employment value and J_U is the firm's. Summing the Bellman equations of both parties yields a recursion for the surplus that is independent of the wage:

$$(r + \nu + \lambda_U(z))\, S_U(x, p, z) = P_U(z)\, x\, p - (r + \nu)\, U_U(x, z) + \lambda_U(z) \int_{p_U^*}^{1} S_U(x, p', z)\, dG_z(p'). \quad (6)$$

The surplus equals discounted output minus the flow value of the worker's outside option, plus a continuation term reflecting the option value of surviving a damage shock and drawing a quality above the reservation threshold.

Reservation quality

The threshold $p_U^*(x, z)$ is defined by $S_U(x, p_U^*, z) = 0$. Higher-ability workers tolerate lower match quality because $P_U(z)\, x\, p$ rises faster in p when x is large. An increase in the outside option—whether through better search conditions or a more attractive training option—raises the threshold and makes fragile matches more likely to dissolve.

Wages

The Nash sharing rule gives the wage

$$w_U(x, p, z) = \beta_U\, P_U(z)\, x\, p + (1 - \beta_U)(r + \nu)\, U_U(x, z). \quad (7)$$

The worker receives a share β_U of flow output plus a flow return proportional to the outside option. Because both terms are linear in p, the wage inherits the linearity of the production function.

Vacancies and free entry

Posting a vacancy costs $k_U(z)$ per unit time and generates contacts at rate $q_U(\theta_U(z))$. Since every new match starts at $p_0 = 1$, free entry requires

$$\frac{k_U(z)}{q_U(\theta_U(z))} = \int_0^1 (1 - \beta_U)\, S_U(x, 1, z)\, \hat{u}_U(x, z)\, dx, \tag{8}$$

where $\hat{u}_U(x, z) = u_U(x, z)/\int u_U(x', z)\, dx'$ is the composition of the unskilled unemployed pool. Because the training cutoff determines which types remain in this pool, the education margin feeds directly into vacancy creation.

3.6 Skilled Labour Market

The skilled market features a job ladder: employed workers search on the job and climb toward better matches. Wages are determined by Nash bargaining in the presence of on-the-job search, following Cahuc et al. (2006).

Match quality and shocks

Skilled matches face two sources of destruction. Exogenous separation occurs at rate $\xi_S(z)$, and match quality is hit by shocks at rate $\lambda_S(z)$. After being shocked, match quality is redrawn from a Beta$(a_\Gamma(z), b_\Gamma(z))$ distribution Γ_z on $[0, 1]$ with density γ_z. If the new draw falls below the reservation cutoff $p_S^*(x, z)$, the match is endogenously destroyed. Unlike in the unskilled market, skilled matches can improve or worsen after a shock—a feature that interacts with the job ladder to generate rich mobility dynamics.

On-the-job search

An employed skilled worker can search on the job at flow cost $\sigma_S \geq 0$. Outside offers arrive at rate $\kappa_S(z)$—the same contact rate that skilled unemployed workers face—with each offer an independent draw $\tilde{p} \sim \Gamma_z$. The worker moves to the outside job if and only if it yields a higher value. Workers in weak matches gain the most from search because the expected improvement is large; workers in strong matches may find the flow cost not worth paying. This generates a second cutoff $p_S^{oj}(x, z) \geq p_S^*(x, z)$: workers with $p < p_S^{oj}$ search on the job, workers with $p \geq p_S^{oj}$ do not.

Surplus

For each search regime $s \in \{0,1\}$, define the total surplus as $S_S^s(x,p,z) = [E_S^s(x,p,z) - U_S(x,z)] + J_S^s(x,p,z)$, and let $S_S(x,p,z) = \max\{S_S^0(x,p,z),\ S_S^1(x,p,z)\}$ be the surplus under the optimal search decision.

For a worker who does not search on the job ($s = 0$), the surplus satisfies

$$(r + \nu + \xi_S(z) + \lambda_S(z))\, S_S^0(x,p,z) = P_S(z)\, x\, p - (r+\nu)\, U_S(x,z) + \big(\lambda_S(z) - \beta_S\, \kappa_S(z)\big) \int_{p_S^*}^1 S_S(x,p',z)\, d\Gamma_z(p'). \tag{9}$$

The coefficient $\lambda_S(z) - \beta_S\, \kappa_S(z)$ on the integral reflects two forces: quality shocks at rate $\lambda_S(z)$ redraw the match and deliver continuation value, while the worker's outside option embeds the contact rate $\kappa_S(z)$ weighted by the bargaining share β_S.

For a searching worker ($s = 1$), the surplus acquires an additional term from the job ladder:

$$\begin{aligned}\big(r + \nu + \xi_S(z) + \lambda_S(z) + \kappa_S(z)(1 - \Gamma_z(p))\big)\, S_S^1(x,p,z) \\ = P_S(z)\, x\, p - \sigma_S - (r+\nu)\, U_S(x,z) \\ + \lambda_S(z) \int_{p_S^*}^1 S_S(x,p',z)\, d\Gamma_z(p') - \beta_S\, \kappa_S(z) \int_{p_S^*}^p S_S(x,p',z)\, d\Gamma_z(p').\end{aligned} \tag{10}$$

The left-hand side includes the poaching hazard $\kappa_S(z)(1-\Gamma_z(p))$: when the worker searches, outside offers that beat the current match quality arrive at rate $\kappa_S(z)$ and are accepted with probability $1 - \Gamma_z(p)$. On the right, the integral from p_S^* to p reflects the worker's partial appropriation of the ladder option through bargaining.

The value of skilled unemployment satisfies

$$(r+\nu)\, U_S(x,z) = b_S + \kappa_S(z) \int_{p_S^*}^1 \beta_S\, S_S(x,\tilde{p},z)\, d\Gamma_z(\tilde{p}). \tag{11}$$

Wage schedules

Nash bargaining with worker share β_S applies in the skilled market as well. The resulting wages differ depending on whether the worker searches. For a non-searching worker,

$$w_S^0(x,p,z) = \beta_S\, P_S(z)\, x\, p + (1-\beta_S)\, b_S + \beta_S(1-\beta_S)\, \kappa_S(z) \int_{p_S^*}^{1} S_S(x,p',z)\, d\Gamma_z(p'). \tag{12}$$

For a searching worker,

$$w_S^1(x,p,z) = \beta_S\, P_S(z)\, x\, p + (1-\beta_S)(b_S + \sigma_S) + \beta_S(1-\beta_S)\, \kappa_S(z) \int_{p_S^*}^{p} S_S(x,p',z)\, d\Gamma_z(p'). \tag{13}$$

Workers who search on the job earn strictly less than non-searchers at the same (x,p). Two forces drive the discount: Nash bargaining shifts the flow search cost partly onto the worker, and firms anticipate that a searching worker is more likely to be poached, reducing the expected duration of the match and thereby the surplus the firm is willing to share. Both effects are strongest near the reservation threshold, where the gain from climbing the ladder is largest.

Vacancies and free entry

Posting a skilled vacancy costs $k_S(z)$ per unit time and generates contacts at rate $q_S(\theta_S(z))$. The pool of seekers includes both unemployed and on-the-job searchers, so the expected value of a contact depends on whom the firm meets. Let $\bar{J}_S(z)$ denote this expected value, averaged over the composition of seekers. Free entry requires

$$k_S(z) = q_S(\theta_S(z))\, \bar{J}_S(z), \tag{14}$$

pinning down $\theta_S(z)$. Because the training cutoff $\bar{x}(z)$ determines the inflow into skilled unemployment and the on-the-job search cutoff determines how many employed workers are active on the market, both policy margins feed into the free-entry condition.

3.7 Stationary Equilibrium

Fix a regime $z \in \mathcal{Z}$. A stationary equilibrium consists of value functions $(U_U,\, U_U^{\text{search}},\, T,\, E_U,\, U_S,\, E_S^0,\, E_S^1)$, firm values $(J_U,\, J_S^0,\, J_S^1)$, policy rules $(\tau,\, p_U^*,\, p_S^*,\, p_S^{oj},\, s^*)$,

market tightness (θ_U, θ_S), and cross-sectional distributions (u_U, t, u_S, e_S) such that:

(i) **Optimality.** Worker and firm value functions satisfy their Bellman equations; training, separation, and on-the-job search rules are optimal given those values.

(ii) **Wage determination.** Wages satisfy the Nash-bargaining rules in equations (7), (12), and (13).

(iii) **Free entry.** Tightness satisfies the zero-profit conditions (8) and (14).

(iv) **Stationarity.** Cross-sectional distributions satisfy flow-balance conditions, with inflows equal to outflows in every state.

(v) **Consistency across markets.** The completion flow from training equals the inflow into skilled unemployment.

The two markets are linked through the training cutoff: it depends on the value of skilled unemployment, which depends on skilled-market tightness, which depends on the distribution of skilled workers, which depends on the training inflow. I solve for the equilibrium with a nested fixed-point algorithm described in Appendix **??**.

3.8 Model Properties

The structure delivers several properties that guide the empirical and policy analysis below.

Training cutoff. The cutoff $\bar{x}(z)$ responds to aggregate conditions through three channels: changes in the skilled wage premium, changes in the probability of finding a skilled job after training, and changes in the value of the unskilled outside option. A regime that raises $P_S(z)$ relative to $P_U(z)$ lowers the cutoff and pushes more workers into training; a regime that raises the skilled contact rate $\kappa_S(z)$ has the same effect by increasing the arrival rate of skilled jobs.

Endogenous separation. Higher-ability workers tolerate lower match quality in both markets, because the flow surplus $P_j(z)\, x\, p$ rises faster in p when x is large. An increase in the outside option—whether through improved search conditions or a more attractive training option—raises reservation thresholds and makes fragile matches more likely to dissolve.

On-the-job search and wage dispersion. The search cutoff $p_S^{oj}(x, z)$ generates wage dispersion in the skilled market even among workers of identical ability. Workers who have

been lucky enough to climb to high-quality matches earn more than those still searching, purely because of the frictional process of job-ladder climbing.

The aggregate state. Because z shifts $\lambda_S(z)$, $\xi_S(z)$, and the match-quality distribution Γ_z jointly with $P_S(z)$, a recession is not merely a productivity shortfall: it also changes the pace at which workers are displaced and the rate at which match quality deteriorates. Separating these channels is the main motivation for the structural approach in Section 4.

4 Data and Estimation

4.1 Data

The model is estimated on U.S. data from four sources. Labour-market stocks and worker transition rates are constructed from the Current Population Survey—Basic Monthly Files (CPS). Wage moments are drawn from the CPS Annual Social and Economic Supplement (ASEC). The skilled employer-to-employer transition rate comes from the Census Bureau's Job-to-Job Flows (J2J) data, which provides quarterly counts of hires from other employers by education group. Vacancy data for market tightness come from the Job Openings and Labor Turnover Survey (JOLTS).

The sample consists of people (civilians, military was excluded) aged 25–54 who report positive labour-force attachment. I partition workers into two skill groups: those without a bachelor's degree (unskilled) and those with a bachelor's degree or above (skilled). This partition is the empirical counterpart of the model's two-market structure. Monthly transition rates—unemployment to employment (UE), employment to unemployment (EU), and employer to employer (EE)—are constructed from matched CPS records using standard methods.

Moments are computed over four windows, each chosen to approximate a stationary regime:

Table 1: Estimation windows

Period	Dates	Length
Baseline fc	Jan 2003 – Dec 2007	60 months
Crisis fc	Jan 2008 – Jun 2009	18 months
Baseline covid	Jan 2015 – Dec 2019	60 months
Crisis covid	Mar 2020 – Dec 2021	22 months

The two baseline windows cover the expansion phases preceding each crisis. The crisis windows cover the contraction through the trough. Within each window, I average the

monthly observations to obtain a single vector of target moments and treat this average as the stationary equilibrium that the model should match.

4.2 Moments and Identification

I estimate the model by the Simulated Method of Moments in a standard fashion. The estimation targets 23 moments grouped into four blocks. Table 2 lists each moment with its model counterpart.

Table 2: Targeted moments

Moment	Model counterpart	Source
Labour-market stocks		
Skilled share	$\int_{\bar{x}}^{1} [u_S(x) + \int e_S(x,p)\,dp]\,dx$	CPS
Training share	$\int_{\bar{x}}^{1} t(x)\,dx$	CPS
Variance of unskilled wages	$\mathrm{Var}[w_U]$	CPS ASEC
Skewness of unskilled wages	$\mathrm{CM}_3[w_U]$	CPS ASEC
Variance of skilled wages	$\mathrm{Var}[w_S]$	CPS ASEC
Skewness of skilled wages	$\mathrm{CM}_3[w_S]$	CPS ASEC
Transition rates		
Job-finding rate, unskilled	$\kappa_U(z)\,[1 - G_z(p_U^*)]$	CPS
Separation rate, unskilled	$\lambda_U(z)\,G_z(p_U^*)$	CPS
Job-finding rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^*)]$	CPS
Separation rate, skilled	$\xi_S(z) + \lambda_S(z)\,\Gamma_z(p_S^*)$	CPS
EE rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^{oj})]$	J2J
Wages		
Mean wage, unskilled	$\mathbb{E}[w_U(x,p,z)]$	CPS ASEC
Mean wage, skilled	$\mathbb{E}[w_S(x,p,z)]$	CPS ASEC
25th percentile wage, unskilled	$Q_{0.25}[w_U]$	CPS ASEC
25th percentile wage, skilled	$Q_{0.25}[w_S]$	CPS ASEC
Median wage, unskilled	$Q_{0.50}[w_U]$	CPS ASEC
Median wage, skilled	$Q_{0.50}[w_S]$	CPS ASEC
Log skill premium	$\mathbb{E}[\ln w_S] - \mathbb{E}[\ln w_U]$	CPS ASEC
Tightness		
θ_U	V_U/U_U	JOLTS / CPS
θ_S	V_S/U_S^{eff}	JOLTS / CPS

The choice of moments reflects the structure of the model. The skilled share and training share directly discipline the training cutoff $\bar{x}(z)$. The cross-sectional wage dispersion moments—variance and skewness of the wage distribution within each skill group—pin down the shape parameters of the match-quality distributions and help separate the ability distribution from the match-quality distribution. Without these moments, the two sources of wage variation would be poorly identified separately.

Transition rates play the central role in identifying market-specific frictions. The unskilled UE rate identifies the contact rate $\kappa_U(z)$, and therefore—through the matching function and free-entry condition—the matching efficiency and vacancy cost. The unskilled EU rate identifies the shock arrival rate $\lambda_U(z)$ and the reservation quality $p_U^*(x, z)$: given the production function, these two objects are tightly linked. On the skilled side, the same logic applies, with the additional information from the employer-to-employer (EE) transition rate. This moment is particularly valuable because it identifies the intensity of on-the-job search: it pins down the contact rate $\kappa_S(z)$ jointly with the on-the-job search cutoff $p_S^{oj}(x, z)$ and helps separate the offer distribution Γ_z from the separation threshold $p_S^*(x, z)$.

Wage levels identify the productivity shifters $P_U(z)$ and $P_S(z)$. The log skill premium identifies the return to training. Wage quantiles—particularly the gap between the 25th percentile and the median—are informative about the shape of the match-quality distributions and the bargaining shares β_j: a compressed lower tail relative to the median suggests a tight match-quality distribution or high bargaining power, while a long lower tail suggests the opposite.

Market tightness $\theta_j(z)$ directly constrains vacancy costs $k_j(z)$ through the free-entry conditions. Without these moments, vacancy costs would be identified only indirectly through their effect on contact rates, which also depend on matching efficiencies.

4.3 Estimation

The SMM estimator minimises the weighted distance between the data moments $\hat{m}$ and their model counterparts $m(\theta)$:

$$\hat{\theta} = \arg\min_{\theta} \big[\hat{m} - m(\theta)\big]' \, W \, \big[\hat{m} - m(\theta)\big], \tag{15}$$

where W is the optimal weight matrix, constructed as the inverse of the estimated variance–covariance matrix of $\hat{m}$. I estimate $\hat{\Sigma}$ using influence-function standard errors with shrinkage applied to off-diagonal elements when the matrix is ill-conditioned; details are in Appendix **??**.

The model moments $m(\theta)$ are computed by solving for the stationary equilibrium at parameter vector θ and evaluating the equilibrium objects described in Section 3.

Parameter classification

Three parameters—the discount rate r, the demographic exit rate ν, and the training completion rate ϕ—are calibrated externally and held fixed across all estimation windows. Table 3

reports their values.

Table 3: Externally calibrated parameters

Parameter	Source	Value
r	Monthly discount rate ($\approx 5\%$ annual)	0.0042
ν	CPS matched-panel labour-force exit hazard	0.0384
ϕ	National Student Clearinghouse completion rate	0.0222

A second group of parameters is fixed externally to simplify the model. Firms and workers share surplus equally and the matching function depends symmetrically on unemployment and vacancies, so the bargaining shares (β_U, β_S) and matching elasticities (η_U, η_S) are all set to 0.5. The OJS flow cost is likewise fixed, at $\sigma_S = 1.1$.

The remaining parameters are estimated, and fall into two groups. *Deep structural parameters* reflect preferences, institutions, and technology that do not change with the aggregate state: the ability distribution (a_ℓ, b_ℓ), the training cost c, the flow payoffs (b_U, b_T, b_S), and the matching efficiencies (μ_U, μ_S). *Regime-specific parameters* capture features of the aggregate state that shift between regimes: the sector productivities (P_U, P_S), the match-quality distribution parameters $(\alpha_U, a_\Gamma, b_\Gamma)$, the vacancy costs (k_U, k_S), the shock arrival rates (λ_U, λ_S), and the exogenous skilled separation hazard ξ_S.

Multi-regime estimation strategy

The estimation proceeds in two stages. In the first stage, I estimate the model on the baseline window, treating all deep structural and regime-specific parameters as free. This stage identifies the full parameter vector from a single cross-section of moments.

In the second stage, I fix the deep structural parameters at their baseline estimates and re-estimate only the regime-specific parameters on the crisis window. This procedure identifies what changed during the crisis: which parameters shifted, by how much, and in which direction. The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the baseline and the crisis—only the aggregate state did.

The same two-stage procedure is applied separately to the financial-crisis moments pair (baseline fc $\rightarrow$ crisis fc) and the COVID pair (baseline covid $\rightarrow$ crisis covid). Comparing the deep structural estimates across the two baselines provides a structural stability check: if the deep parameters are similar across the two expansions, the identifying assumption that they are regime-invariant is internally consistent.

5 Estimation Results

This section presents the estimation results. I begin with the baseline equilibrium estimated on the pre-crisis expansion (2003–2007), examine the model's internal structure by simulating it for the estimated parameters, and assess the quality of the moment fit. I then report the parameter estimates and turn to the crisis re-estimation—first for the financial crisis (2008–2009), then for COVID (2020–2021)—to identify which features of the aggregate state shifted during each downturn and how those shifts affected the training margin.

5.1 Baseline Equilibrium

The model is estimated on the pre-financial-crisis baseline window (2003–2007) and solved at the estimated parameters. I describe the resulting equilibrium objects in turn.

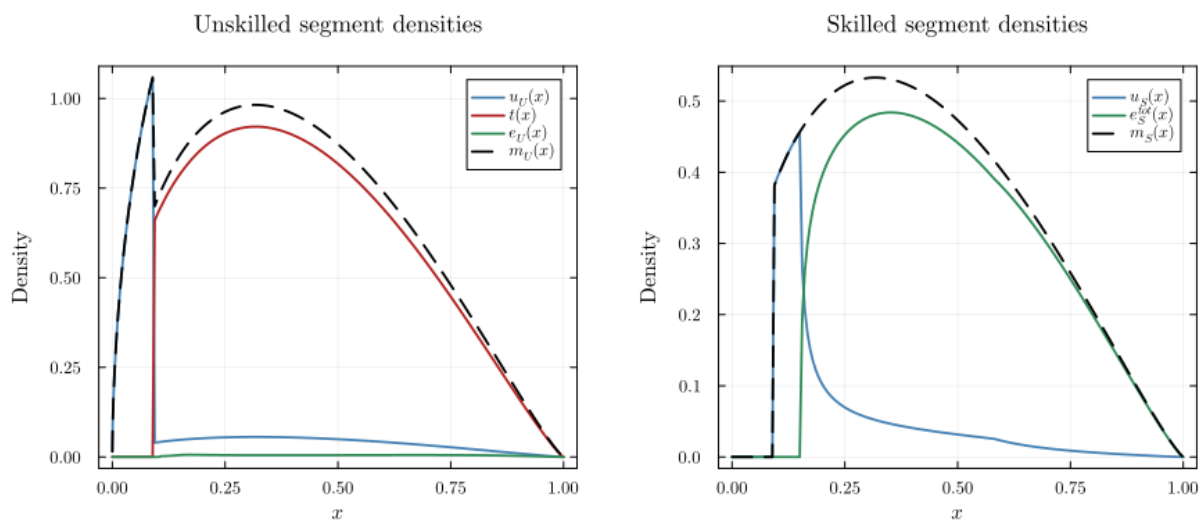

Figure 1: Cross-sectional densities over worker type x: unskilled segment (left) and skilled segment (right). The training cutoff $\bar{x}$ partitions the two markets.

Cross-sectional distributions. The figure shows the endogenous segmentation of the economy. The training cutoff partitions the ability distribution into two pools with different mean characteristics. Above $\bar{x}$, the mass flows through the training pipeline into skilled unemployment and then into skilled employment. Below $\bar{x}$, workers remain in the unskilled market permanently.

Training decision. For low-ability workers, the value of unskilled search exceeds the net value of training: the training cost is too high and the expected gains from the skilled market too low. For high-ability workers, the net value of training dominates: training costs are low,

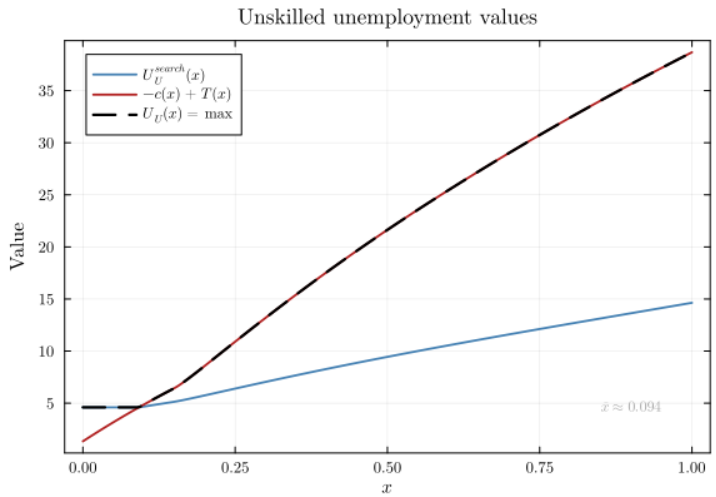

Figure 2: Unskilled unemployment values

and the continuation value of the skilled market—including job-ladder gains from on-the-job search—is high. The gap between the two curves at $\bar{x}$ is zero by construction; above $\bar{x}$ it widens, reflecting the increasing net gain from training for higher-ability types.

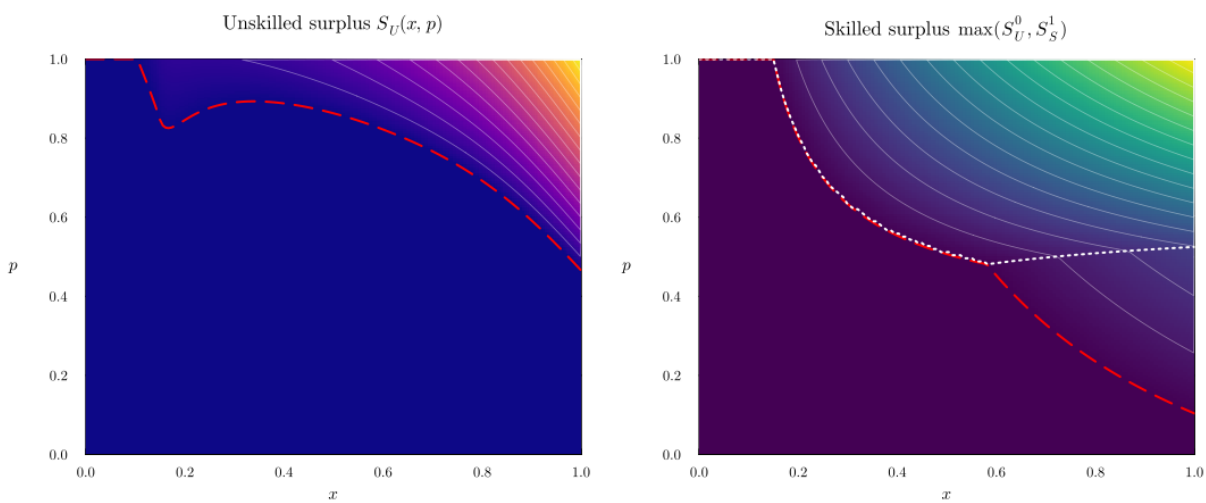

Figure 3: Match surplus over the (x, p) plane: unskilled surplus $S_U(x, p)$ (left) and skilled surplus $\max(S_S^0, S_S^1)$ (right). Overlaid curves mark the reservation quality $p_j^*(x)$ and, in the skilled panel, the on-the-job search cutoff $p_S^{oj}(x)$.

Surplus and separation rules. The surplus maps summarise the equilibrium in each market. Sorting is present in both markets. A key comparative static is the slope of the reservation cutoff in x: flatter curves mean that the training threshold has a weaker effect on match stability.

5.2 Moment Fit

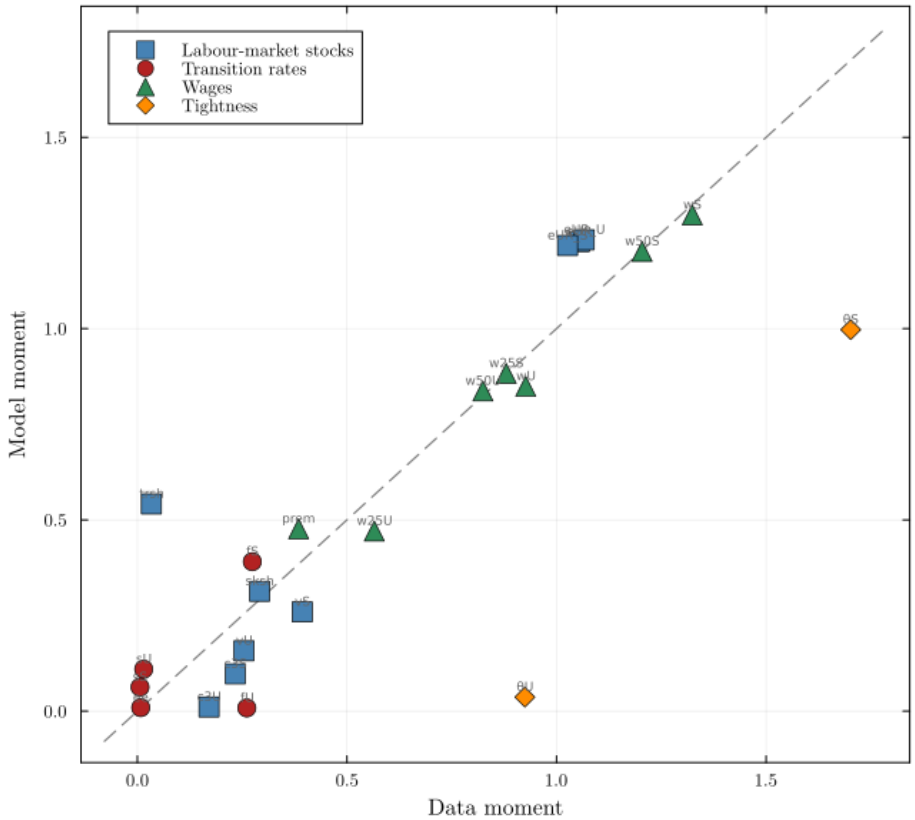

Figure 4: Model fit: data moments versus model counterparts for the pre-financial-crisis baseline.

Tables 4 and 5 report the model fit for both baseline windows—the pre-FC expansion (2003–2007) and the pre-COVID expansion (2015–2019)—placing data moments alongside their model counterparts and the difference between them. The two baselines are shown side by side to facilitate the structural stability comparison: the deep parameters are estimated on the FC baseline and held fixed when the model is re-solved on the COVID baseline, so the COVID column reveals how well the same preference, technology, and institutional parameters rationalise a second, independent expansion twelve years later.

Wages. The wage block is the best-matched component of the estimation. For the FC baseline, mean wages, medians, and 25th-percentile wages are all within 1–9 percentage points of their data targets, and several—particularly the median and 25th-percentile of skilled wages—are matched almost exactly (differences of -0.003 and -0.002, respectively).

Table 4: Model Fit (A): Labour-Market Stocks and Transition Rates

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Labour-market stocks						
exp(UR total)	1.0543	1.2274	+0.1731	1.0462	1.3041	+0.2579
	(0.0057)	(0.0025)		(0.0066)	(0.0019)	
exp(UR unskilled)	1.0661	1.2321	+0.1661	1.0582	1.3648	+0.3066
	(0.0063)	(0.0025)		(0.0086)	(0.0026)	
exp(UR skilled)	1.0264	1.2171	+0.1907	1.0250	1.1496	+0.1246
	(0.0043)	(0.0026)		(0.0027)	(0.0013)	
Skilled share	0.2915	0.3133	+0.0218	0.3596	0.2652	-0.0945
	(0.0208)	(0.0011)		(0.0295)	(0.0007)	
Training share	0.0334	0.5416	+0.5081	0.0450	0.4584	+0.4134
	(0.1061)	(0.0018)		(0.0903)	(0.0011)	
Var(w) unskilled	0.2543	0.1578	-0.0965	0.2860	0.1555	-0.1305
	(0.0643)	(0.0259)		(0.0432)	(0.0371)	
CM3(w) unskilled	0.1709	0.0110	-0.1599	0.2624	0.0402	-0.2222
	(0.0740)	(0.0124)		(0.0454)	(0.0394)	
Var(w) skilled	0.3934	0.2611	-0.1323	0.4498	0.3023	-0.1475
	(0.0477)	(0.0315)		(0.0480)	(0.0402)	
CM3(w) skilled	0.2337	0.0980	-0.1357	0.3514	0.1400	-0.2114
	(0.0470)	(0.0359)		(0.0320)	(0.0272)	
Transition rates						
JFR unskilled	0.2610	0.0083	-0.2528	0.2565	0.0639	-0.1926
	(0.0849)	(0.0177)		(0.0846)	(0.0099)	
Sep. rate unskilled	0.0151	0.1097	+0.0945	0.0132	0.0241	+0.0109
	(0.0634)	(0.0384)		(0.1165)	(0.0155)	
JFR skilled	0.2741	0.3907	+0.1166	0.2717	0.4577	+0.1860
	(0.1500)	(0.0704)		(0.1398)	(0.0956)	
Sep. rate skilled	0.0062	0.0628	+0.0566	0.0059	0.0428	+0.0369
	(0.1246)	(0.0727)		(0.1617)	(0.0326)	
EE rate skilled	0.0080	0.0052	-0.0027	0.0085	0.0032	-0.0053
	(0.0624)	(0.0624)		(0.0366)	(0.0366)	

Table 5: Model Fit (B): Wages and Tightness

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Wages						
Mean wage unskilled	0.9265	0.8493	-0.0772	0.9166	0.8030	-0.1136
	(0.0360)	(0.0178)		(0.0315)	(0.0269)	
Mean wage skilled	1.3239	1.2951	-0.0289	1.3144	1.1705	-0.1440
	(0.0346)	(0.0208)		(0.0289)	(0.0258)	
p25 wage unskilled	0.5652	0.4709	-0.0943	0.5482	0.4779	-0.0703
	(0.0393)	(0.0190)		(0.0453)	(0.0392)	
p25 wage skilled	0.8800	0.8776	-0.0024	0.8411	0.7201	-0.1210
	(0.0392)	(0.0231)		(0.0278)	(0.0264)	
p50 wage unskilled	0.8244	0.8371	+0.0127	0.7920	0.7337	-0.0583
	(0.0384)	(0.0194)		(0.0338)	(0.0285)	
p50 wage skilled	1.2042	1.2008	-0.0033	1.1594	1.0480	-0.1114
	(0.0377)	(0.0237)		(0.0298)	(0.0257)	
Skill premium	0.3846	0.4743	+0.0897	0.3851	0.3959	+0.0108
	(0.0136)	(0.0113)		(0.0172)	(0.0141)	
Tightness						
θ_U	0.9244	0.0367	-0.8878	1.5849	5.6596	+4.0747
	(0.2101)	(0.0880)		(0.2524)	(0.0567)	
θ_S	1.7020	0.9972	-0.7048	2.8943	1.3089	-1.5854
	(0.2508)	(0.1374)		(0.1508)	(0.0555)	

The skill premium is slightly over-predicted in the FC window (+0.09) but nearly exact in the COVID window (+0.01). The COVID baseline shows somewhat larger deviations in skilled wage levels (mean wage skilled: −0.14; median skilled: −0.11), suggesting that the deep parameters estimated on the FC expansion generate a slightly compressed skilled wage distribution when applied to the tighter 2015–2019 labour market. The unskilled wage moments degrade less across baselines, consistent with the simpler structure of the unskilled block, where matches start at frontier quality and only deteriorate.

Labour-market stocks. Unemployment rates in exponentiated form are over-predicted in both baselines, with the discrepancy larger for the COVID window (total: +0.26 vs. +0.17; unskilled: +0.31 vs. +0.17). The skilled unemployment rate, however, is closer to the data in the COVID baseline (+0.12) than in the FC baseline (+0.19), indicating that the model captures the differential tightening of the skilled market during the later expansion more effectively than it captures the corresponding unskilled dynamics.

The skilled share is well matched in the FC baseline (+0.02) but under-predicted for the COVID baseline (−0.09). This is consistent with the secular rise in educational attainment between the two periods—a trend the model cannot generate, since the deep parameters governing the type distribution (a, b) and training cost c are held fixed at FC baseline values. A time-varying composition shift in the population, operating outside the model's regime-specific channels, likely accounts for the remainder.

The training share is the largest point of disagreement in both baselines: the model generates a share of approximately 0.46–0.54, while the data—based on CPS college-enrolment flags for individuals without a BA—shows only 0.03–0.05. The data standard error on this moment is itself very large (0.09–0.11), reflecting the difficulty of measuring training stocks from cross-sectional survey indicators. Three factors contribute to the discrepancy. First, the model's training state captures all workers who have initiated the pipeline toward the skilled market, regardless of whether they would be classified as "enrolled in college" by the CPS. Second, the CPS measure excludes part-time and non-degree enrolments that may still represent human-capital investment in the model's sense. Third, with only a single training-cost parameter c disciplining this moment, the estimator trades off training-share fit against the other 22 moments in the objective, and the large data variance reduces the weight the optimal matrix places on this moment. The over-prediction should be borne in mind when interpreting the model's training cutoff $\bar{x}(z)$ and the implied mass flowing through the training pipeline.

Wage dispersion. The model systematically under-predicts the variance and third central moment of wages in both skill groups and both baselines. The under-prediction is more pronounced for the third central moment (skewness), particularly in the COVID baseline (CM3 skilled: -0.21; CM3 unskilled: -0.22). In the model, within-group wage dispersion arises from two sources: heterogeneity in innate type x and dispersion in match quality p. The compressed model variance suggests that the estimated match-quality distributions are tighter than what the data warrant, or that the model lacks an additional source of residual wage dispersion—such as within-match human capital accumulation or firm heterogeneity—that would fatten the tails of the cross-sectional wage distribution. The gap between data and model skewness, in particular, points toward a missing right tail: the model cannot generate the long upper tail of the wage distribution without allowing for firm-side heterogeneity or on-the-job skill growth, both of which are abstracted away.

Transition rates. The transition-rate block presents the most challenging fit. The unskilled job-finding rate (JFR) is severely under-predicted in both baselines (FC: 0.008 vs. 0.261; COVID: 0.064 vs. 0.257), indicating that the model's equilibrium unskilled contact rate—governed by matching efficiency μ_U, tightness θ_U, and the matching elasticity—is too low. Conversely, the unskilled separation rate is over-predicted in the FC baseline (+0.09) but nearly matched during the COVID baseline (+0.01), suggesting that the estimated shock-arrival rate λ_U and reservation rule $\underline{p}_U(x, z)$ are reasonable for the later expansion but generate excessive turnover when applied to the earlier one.

In the skilled market, the pattern reverses: the JFR is over-predicted (+0.12 FC; +0.19 COVID), while the separation rate is over-predicted by a smaller margin (+0.06 FC; +0.04 COVID). The skilled EE rate—the moment that identifies on-the-job search intensity—is well matched in both windows (differences of -0.003 and -0.005), confirming that the job-ladder mechanism and the OJS cutoff $p_S^{oj}(x, z)$ are appropriately disciplined. The opposing signs of the JFR discrepancies across the two markets reflect a tension inherent in the model: with matching efficiencies μ_U, μ_S held fixed across regimes, the estimator cannot simultaneously generate high contact rates in one market and moderate contact rates in the other without adjusting tightness, which is itself constrained by the free-entry conditions.

Market tightness. Tightness is the least well matched set of moments. In the FC baseline, both θ_U and θ_S are under-predicted (-0.89 and -0.70, respectively), whereas in the COVID baseline θ_U is dramatically over-predicted (+4.07) while θ_S remains under-predicted (-1.59). The sign reversal for unskilled tightness across the two baselines is notable: the FC deep parameters imply vacancy costs and matching efficiency levels that, when confronted with the

much tighter 2015–2019 labour market, generate an equilibrium with far too many vacancies relative to the unskilled unemployed pool. This is consistent with the secular decline in the matching function's efficiency documented in the macro-labour literature, a time trend that the model—which treats μ_U as a fixed deep parameter—cannot accommodate. The discrepancy in skilled tightness is smaller in relative terms and consistent in sign across baselines, suggesting that the skilled free-entry condition is better anchored.

Cross-baseline assessment. Taken together, the moment fit delivers a clear hierarchy: wages are matched tightly, stock moments are matched at an intermediate level, and transition rates and tightness are matched least precisely. This ordering is consistent with the structure of the estimation—wage moments are pinned down by productivity shifters P_U, P_S with relatively few competing parameters, while transition rates depend on the interaction of tightness, matching efficiency, shock arrival, and endogenous separation, creating more scope for trade-offs in the objective function. The COVID baseline shows somewhat larger deviations than the FC baseline on average, which is expected: the deep parameters were estimated to fit the FC expansion and are only held fixed—not re-optimised—for the COVID window. That the wage fit remains strong and the skilled EE rate remains accurate across both baselines provides the core evidence for the structural stability of the deep parameters, supporting the identifying assumption that preferences and matching technology did not change between the two expansions.

5.3 Parameter Estimates

5.4 Crisis Assessment

The second-stage estimation fixes the deep structural parameters at their baseline values and re-estimates the ten regime-specific parameters on each crisis window. The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the expansion and the recession—only the aggregate state did. Tables 6 and 7 report the full set of estimates across all four windows. I discuss the regime-specific shifts for each crisis in turn, then compare the two episodes.

Productivity shifters. In both crises, unskilled productivity P_U falls while skilled productivity P_S rises. During the financial crisis the unskilled decline is moderate (-0.13, or -6.6%), while during COVID it is much larger (-0.64, or -23.5%). The rise in P_S is likewise more pronounced in COVID ($+1.00$, or $+25.4\%$) than in the financial crisis ($+0.30$, or $+7.9\%$). Because P_U and P_S are identified primarily by wage levels, the divergence reflects

Table 6: Parameter Estimates (A): Calibrated and Deep Structural

		Financial Crisis			COVID		
Parameter	Description	Baseline	Crisis	Δ	Baseline	Crisis	Δ
Externally calibrated							
r	Discount rate	0.00417	0.00417	—	0.00417	0.00417	—
ν	Demographic exit rate	0.03841	0.03841	—	0.03841	0.03841	—
φ	Training completion rate	0.02222	0.02222	—	0.02222	0.02222	—
Deep structural							
a_ℓ	Worker type shape	1.50538	1.50538	—	1.06876	1.50538	—
b_ℓ	Worker type shape	2.87219	2.87219	—	2.56120	2.87219	—
c	Training cost coeff.	2.94357	2.94357	—	2.91536	2.94357	—
b_U	Unskilled UI flow	0.19600	0.19600	—	0.23613	0.19600	—
b_T	Training flow	1.02176	1.02176	—	1.11815	1.02176	—
b_S	Skilled UI flow	0.56562	0.56562	—	0.43460	0.56562	—
μ_U	Unskilled matching eff.	0.04330	0.04330	—	0.02687	0.04330	—
η_U	Unskilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_U	Unskilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
μ_S	Skilled matching eff.	1.18859	1.18859	—	0.84933	1.18859	—
η_S	Skilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_S	Skilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
σ_S	OJS flow cost	1.10000	1.10000	—	1.10000	1.10000	—

Table 7: Parameter Estimates (B): Regime-Specific

		Financial Crisis			COVID		
Parameter	Description	Baseline	Crisis	Δ	Baseline	Crisis	Δ
Regime-specific							
P_U	Unskilled productivity	2.04377	1.90910	-0.13466	2.73686	2.09363	-0.64323
P_S	Skilled productivity	3.75855	4.05600	+0.29746	3.95147	4.95622	+1.00476
α_U	Unskilled damage shape	5.89456	5.94021	+0.04565	5.43224	7.78774	+2.35551
a_Γ	Skilled offer shape	2.75946	2.91353	+0.15407	3.39435	2.14112	-1.25323
b_Γ	Skilled offer shape	0.70915	1.14897	+0.43981	1.00198	0.82445	-0.17753
k_U	Unskilled vacancy cost	0.00979	0.00351	-0.00628	0.00207	0.00100	-0.00107
λ_U	Unskilled damage rate	0.29197	0.18142	-0.11054	0.32073	0.92715	+0.60642
k_S	Skilled vacancy cost	0.17944	0.19874	+0.01930	0.16986	0.92440	+0.75454
ξ_S	Skilled exog. sep. rate	0.00000	0.00000	+0.00000	0.00000	0.00000	-0.00000
λ_S	Skilled quality shock	0.42931	0.38727	-0.04204	0.43895	0.49556	+0.05661

the asymmetric impact of each crisis on the two sectors. The unskilled market—dominated by service, retail, and construction jobs—bore the direct brunt of both the housing bust and the lockdowns. The skilled market proved more resilient, and the rise in P_S is consistent with the rapid adoption of remote work and digital technologies that raised the marginal product of educated workers during both downturns. The widening gap between P_S and P_U during COVID, in particular, matches the well-documented *K-shaped* recovery in which high-skill sectors rebounded quickly while low-skill sectors stagnated.

Unskilled damage shocks. The two crises generate sharply different patterns in the unskilled shock process. During the financial crisis, the damage arrival rate λ_U *falls* from 0.292 to 0.181 (-38%), while the damage shape α_U is essentially unchanged ($+0.05$). The decline in λ_U is counterintuitive if one expects recessions to increase the frequency of adverse match-specific draws. However, λ_U is jointly identified with the reservation threshold $\underline{p}_U(x, z)$ through the unskilled separation rate: what matters for separations is the product $\lambda_U \cdot G_z(\underline{p}_U)$. A decline in P_U raises the outside-option value relative to match output, pushing up the reservation threshold and increasing the probability of endogenous destruction conditional on any shock. The estimator compensates by lowering λ_U to prevent the separation rate from overshooting its target.

During COVID, the pattern reverses: λ_U nearly triples ($0.321 \to 0.927$, $+189\%$), but α_U rises sharply as well ($5.43 \to 7.79$, $+43\%$). Since post-shock match quality follows a Beta$(\alpha_U, 1)$ distribution, a higher α_U concentrates the post-shock draw closer to 1, making each individual shock less damaging. COVID is thus characterised by *very frequent but individually milder* damage shocks in the unskilled market—consistent with the wave-like pattern of lockdowns and reopenings, in which unskilled matches were repeatedly disrupted but often at reduced intensity (e.g. capacity restrictions rather than full closures).

Skilled quality shocks. The skilled shock arrival rate λ_S moves modestly in both crises: a small decline during the financial crisis (-0.04) and a small increase during COVID ($+0.06$). Neither shift is large enough to indicate a fundamental change in the pace of skilled-match revaluation. This is consistent with the relative insulation of the skilled market from both downturns.

Vacancy costs. The unskilled vacancy cost k_U falls in both crises (FC: -0.006; COVID: -0.001). Although a rise in vacancy costs would be the textbook prediction for a demand-driven recession, the estimated decline can be rationalised through the free-entry condition. During a downturn, the vacancy-filling rate $q_U(\theta_U)$ rises as tightness falls and the pool of unemployed

job seekers expands. If the expected match surplus does not fall proportionally—because, for instance, the reservation threshold adjusts upward and only the most productive matches survive—then free entry can be satisfied at a lower posting cost. The decline in k_U therefore absorbs the compositional improvement in the quality of new hires that accompanies a slack labour market.

The skilled vacancy cost k_S rises in both crises, but the magnitudes differ dramatically. During the financial crisis the increase is negligible (+0.02, or +11%), whereas during COVID it is enormous ($0.170 \to 0.924$, a $4.4\times$ increase). This is the single largest parameter shift in either crisis and captures the unprecedented disruption to skilled hiring during the pandemic. Lockdowns, travel restrictions, and the abrupt transition to remote recruitment raised the effective cost of posting and filling skilled vacancies far beyond anything observed during the financial crisis.

Exogenous skilled separation. The exogenous skilled separation hazard ξ_S remains at zero in all four regimes. The model attributes all skilled-market job destruction to endogenous separation via the reservation rule $\underline{p}_S^*(x, z)$ and quality-shock redraws, never to exogenous layoffs unrelated to match-specific productivity. This is a strong finding: it implies that even during the pandemic—when large-scale layoffs were widespread—the estimated shock process is rich enough to rationalise the observed separation patterns without an additional exogenous margin.

Skilled offer distribution. The skilled offer distribution $\Gamma_z = \text{Beta}(a_\Gamma, b_\Gamma)$ shifts differently across the two crises. During the financial crisis, both parameters increase (a_Γ: +0.15; b_Γ: +0.44), with the rise in b_Γ dominating. Since a higher b_Γ pulls mass toward the lower end of the support, the implied mean offer quality falls from $a/(a+b) = 0.80$ to 0.72, and the distribution becomes more concentrated—the variance of available skilled-match quality narrows. The financial crisis thus degraded the average quality of skilled job offers while compressing their dispersion.

During COVID, both parameters fall (a_Γ: -1.25; b_Γ: -0.18), with the decline in a_Γ dominating. The mean offer quality falls from 0.77 to 0.72—a similar decline as in the FC—but the variance *increases* substantially (from 0.033 to 0.051). The COVID crisis therefore generated a more dispersed distribution of skilled-match quality: some firms offered very good matches (those that thrived during the pandemic), while others offered poor ones (those struggling to adapt), widening the spread of outcomes for trained workers entering the skilled market.

Comparing the two crises. Table 8 summarises the qualitative patterns across the two episodes.

Table 8: Qualitative Comparison of Regime-Specific Shifts

Channel	Financial Crisis	COVID
Unskilled productivity P_U	Moderate decline (-7%)	Large decline (-24%)
Skilled productivity P_S	Moderate rise ($+8\%$)	Large rise ($+25\%$)
Unskilled shock frequency λ_U	Falls (-38%)	Triples ($+189\%$)
Unskilled shock severity α_U	Unchanged	Milder (higher α_U)
Skilled shock frequency λ_S	Small decline	Small rise
Unskilled vacancy cost k_U	Falls (-64%)	Falls (-52%)
Skilled vacancy cost k_S	Small rise ($+11\%$)	Large rise ($+444\%$)
Exogenous skilled separation ξ_S	Zero	Zero
Skilled offer quality (mean)	Falls	Falls
Skilled offer quality (dispersion)	Narrows	Widens

The financial crisis emerges as a relatively symmetric demand shock with moderate parameter shifts: both sector productivities move modestly, shock processes barely change, and vacancy costs adjust by small amounts. The COVID crisis, by contrast, is a far more asymmetric and disruptive event. The gap between skilled and unskilled productivity widens dramatically, unskilled matches are bombarded by frequent (if individually milder) shocks, and the cost of posting skilled vacancies increases more than fourfold. The two crises also differ in what they do to the skilled offer distribution: the financial crisis compressed it, while COVID dispersed it. These differences are economically plausible—the financial crisis was a broad credit-driven contraction that affected most sectors similarly, while the pandemic created sharp winners and losers across industries and occupations, generating the heterogeneity reflected in the parameter estimates.

5.5 Implications for Training

[TODO: Insert the training cutoff values $\bar{x}(z_0)$ and $\bar{x}(z_1)$ for each crisis pair once computed. Compare the baseline and crisis cutoffs: a lower $\bar{x}$ during the crisis means more workers find it optimal to train—the model generates countercyclical enrolment endogenously. Connect to Dellas and Sakellaris (2003), Barr and Turner (2015), and Blom et al. (2021). The mechanism is a decline in the unskilled contact rate $\kappa_U(z)$, which compresses the continuation value of remaining unskilled and makes training relatively more attractive.]

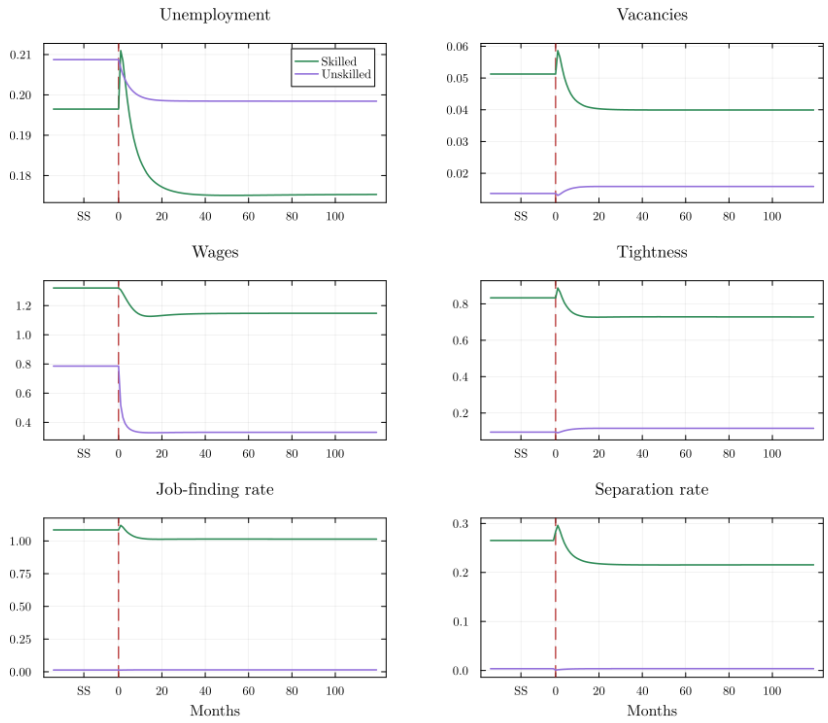

Figure 5: Model transition dynamics after the regime switch.

5.6 Transition Dynamics

6 Education-Subsidy Policies

The preceding sections used the estimated model to interpret two crises. This section turns the model toward policy. Search frictions create a hold-up problem for human-capital investment: because wages are set after training is complete, workers underinvest relative to the social optimum (Acemoglu and Shimer, 1999). The government can correct this distortion by subsidising the training margin, but the design of the subsidy matters. I take one of the estimated baselines, hold all parameters fixed, and ask: how does the stationary equilibrium respond to two alternative education subsidies that target the same margin through different channels?

Policy A: higher training flow payoff b_T. The government provides monthly stipends to workers currently enrolled in training—living-cost grants or tuition waivers that raise disposable income during the training spell. In the model, this is a direct increase in the flow value b_T while the worker occupies the training state. It makes training more attractive by reducing the per-period cost of non-employment during the spell. The transfer is temporary: it stops once training is complete.

Policy B: lower training cost $c(x)$. The government invests in higher-education quality—better equipment, instructors, and facilities—at no additional cost to students. The subjective disutility of studying falls at every ability level. In the model, this is captured as a proportional reduction in the one-time cost function $c(x) = a_c\, e^{-x}$, replacing a_c with $(1-\delta)\, a_c$ for a subsidy rate $\delta \in (0,1)$. The cost reduction is permanent and lowers the up-front hurdle to initiating training.

6.1 Channels and Exercise Design

Both policies shift the training cutoff $\bar{x}(z)$ downward—more workers choose to train—but they operate through different channels. Policy A raises the value of the training state $T(x,z)$ by increasing b_T in equation (4). Because $T(x,z)$ feeds into the overall value of unskilled unemployment through equation (5), the net gain from training $-c(x)+T(x,z)$ rises at every ability level, and the indifference point shifts left. Policy B lowers $c(x)$ directly, making the training option $-c(x)+T(x,z)$ more attractive without changing the value of the training state itself.

The distinction matters for general-equilibrium feedbacks. Under both policies, the lower cutoff sends more workers through the training pipeline into skilled unemployment. The resulting increase in the effective pool of skilled seekers affects skilled-market tightness $\theta_S(z)$ through the free-entry condition (14). Whether tightness rises or falls depends on the vacancy-creation response: if firms post enough additional skilled vacancies to absorb the new entrants, the skilled contact rate $\kappa_S(z)$ may actually increase, reinforcing the subsidy. If not, congestion in the skilled market partially offsets the direct policy effect. This tightness feedback is the key general-equilibrium channel that competitive models of education policy—such as Heckman et al. (1998) and Abbott et al. (2019)—cannot capture.

For each policy, I solve the new stationary equilibrium at three subsidy intensities—10%, 25%, and 50%—holding all other parameters at their estimated baseline values. For Policy A, each intensity corresponds to a percentage increase in b_T; for Policy B, to a percentage reduction in a_c.

6.2 Outcomes

For each policy and intensity level, I report the new stationary-equilibrium values of the training cutoff, the training stock, the skilled share, unemployment rates and market tightness in both sectors, mean wages, transition rates, and worker welfare at various ability levels.

[TODO: Insert results tables and figures here once the policy counterfactuals are computed.]

6.3 Discussion

Several questions guide the interpretation of the results.

Cost-effectiveness. Policy A carries a recurring fiscal cost—stipends paid every period for every trainee—that scales with both the training stock and the duration of the training spell. Policy B carries a one-time capital cost that permanently lowers $c(x)$. Even if both policies shift the cutoff by the same amount, their fiscal implications differ.

Diminishing returns. At high subsidy rates, one policy may saturate faster than the other. If the marginal worker brought into training by a large b_T increase has low ability and therefore a long expected training duration, the per-worker cost of Policy A grows disproportionately. Policy B, by contrast, lowers the cost function uniformly, and its marginal effect on the cutoff may decline more slowly.

General-equilibrium feedbacks. The inflow of trained workers into the skilled market depresses skilled wages and may tighten the skilled market enough to partially offset the subsidy. Whether this dampening is quantitatively important depends on the elasticity of vacancy creation, which the free-entry condition pins down from the estimated parameters.

Baseline dependence. The ranking of the two policies may depend on the current state of the economy—whether the baseline is the pre-financial-crisis or the pre-COVID expansion. This interaction with the regime analysis is not a full regime-change exercise but a comparison of how the same subsidy performs under different initial conditions.

The connection to the empirical programme-evaluation literature is direct. Card et al. (2018) find that training programmes generate larger effects in the medium run than in the short run—a pattern consistent with the job-ladder dynamics in the skilled sector, where newly trained workers enter at the bottom of the match-quality distribution and climb over time. Lindner and Reizer (2020) show that the timing of benefit payments—flow versus frontloaded—generates different search responses holding total transfers constant, providing reduced-form motivation for distinguishing Policy A from Policy B.

7 Conclusion

This paper developed a structural search model in which workers endogenously choose whether to train and enter a higher-skill labour market, and proposed an estimation strategy to bring it to data. The model nests two segmented markets—an unskilled sector with endogenous job destruction and a skilled sector with on-the-job search and a job ladder—linked by a training cutoff that responds to wages, contact rates, and non-employment payoffs in both sectors.

Five features of the framework stand out.

First, training decisions respond endogenously to labour-market conditions, and the direction of the response depends on the nature of the shock. The training cutoff is not a fixed parameter: it shifts with aggregate conditions through the skilled wage premium, the probability of finding a skilled job after training, and the value of the unskilled outside option.

Second, the model generates qualitatively different predictions for different types of crises. A symmetric demand shock raises the option value of waiting and pushes the training threshold up; an asymmetric supply-and-reallocation shock that disproportionately damages the unskilled sector compresses the continuation value of remaining unskilled and lowers the threshold, drawing more workers into training.

Third, the multi-regime estimation strategy—estimating deep parameters on a pre-crisis baseline, fixing them, and re-estimating only regime-specific parameters during crisis—is designed to identify precisely which features of the aggregate state shifted. Comparing deep parameters across two independent baselines provides a structural stability check on the identifying assumption that preferences and technology are regime-invariant.

Fourth, the model's transition dynamics imply that the adjustment of the skilled share is slow, driven by the pipeline delay between training initiation and labour-market entry. Cyclical shocks can therefore generate long-run hysteresis in skill composition even when the shock itself is temporary.

Fifth, education subsidies that target the same margin—the training cutoff—through different channels can produce different outcomes once general-equilibrium feedbacks are accounted for. Raising the training flow payoff (stipends) and lowering the one-time training cost (institutional investment) both shift the cutoff, but they differ in fiscal cost, in how they interact with the training pipeline, and in the tightness response they induce in the skilled market.

The model abstracts from several features that could enrich the analysis. Within-market human capital accumulation—allowing a skill multiplier to evolve with experience—would add a second margin of adjustment that interacts with the training decision. Heterogeneous firms would allow the model to speak to the sorting patterns documented by Lise et al. (2016) and Lise and Robin (2017). Temporary regime changes, rather than permanent switches, would better match the observed recovery dynamics after each crisis. These extensions are left for future work.

The broader message is that structural models which nest regime changes and policy interventions provide a framework for crisis-contingent and investment-oriented labour-market policy. The nature of the crisis determines how workers reallocate across markets, and the design of the subsidy determines whether the reallocation is reinforced or dampened by general-equilibrium feedbacks.

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